

29) Construct a nonzero 3×3 matrix A and a nonzero vector b such that b is in $\text{Nul } A$.

Solution: By Defn, $\text{Nul } A$ represents the set of all vectors x s.t. $Ax=0$.

Let $A = \begin{bmatrix} 1 & 5 & 6 \\ 4 & 2 & 6 \\ 1 & -1 & 0 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ $Ax = 1 \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} + -1 \begin{bmatrix} 6 \\ 6 \\ 0 \end{bmatrix}$
 (By Defn. of Ax)
 $= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\therefore b \in \text{Nul } A \quad \Pi$

30) Suppose the columns of a matrix $A = [a_1 \dots a_p]$ are linearly independent. Explain why $\{a_1, \dots, a_p\}$ is a basis for $\text{Col } A$.

Solution: Let $a_1, \dots, a_p \in \mathbb{R}^m$. $\therefore A$ is a $m \times p$ matrix.
 $\therefore$ The columns of A are linearly independent, the matrix equation $Ax=0$ has only the trivial solution. By Theorem 3 from Chapter 1, this is equivalent to solving the system of equations corresponding to the augmented matrix $[A \ 0]$. By Theorem 2 from Chapter 1, since the solution is unique, each of the p columns of A are pivot columns. (Wrong Solution)

Solution: Let $a_1, \dots, a_p \in \mathbb{R}^m$. By Example 3, the set of all linear combinations of $a_1, \dots, a_p$ is a subspace of $\mathbb{R}^m$. $\therefore \text{Span}\{a_1, \dots, a_p\}$ is a subspace of $\mathbb{R}^m$. $\therefore a_1, \dots, a_p$ are the columns of A , $\therefore \text{Span}\{a_1, \dots, a_p\} = \text{Col } A$. $\therefore \{a_1, \dots, a_p\}$ is a basis for $\text{Col } A$. The last 3 arguments follow from the definition of Span, Column space and Basis respectively. Π

In Exercises 31-36, respond as comprehensively as possible, and justify your answer.

31) Suppose F is a 5×5 matrix whose column space is not equal to $\mathbb{R}^5$. What can you say about $\text{Nul } F$ of F
Solution: By Defn, column space represents the set of all linear combinations of the columns of F . $\therefore$ By Defn, $\text{Col } F = \text{Span}\{\text{columns of } F\}$.